复旦大学计算机科学技术学院

2011-2012 学年第一学期《操作系统》期末考试试卷

A 卷　　共 8 页　　2012 年 1 月

课程代码：COMP130008.01-04　　考试形式：☐开卷　☑闭卷

（本试卷答卷时间为 120 分钟，答案必须写在试卷上，做在草稿纸上无效）

NO aids (books, notes, discussion etc.) are allowed. Ask the instructor if you have any questions.

Student ID_____Name_____Score_____

Question	1	2	3	4	Sum
Score					

（装订线内不要答题）

1. Choice（36%）fill in the blank with ONE most suitable answer

(1) Which of the following statements on memory management are NOT true._____C_____

A. For each address translation, the system with TLB is definitely faster than without TLB.

B. TLB is transparent to application.

C. A TLB miss will definitely result in disk operation to swap in the page

D. A page table is still needed though TLB can achieve fast address translation.

(2) A semaphore s is initialized as 2. After four processes (P1, P2, P3, and P4) have done the following operations sequentially:

(P1, wait(s))→(P2, wait(s))→(P3, wait(s))→(P4, signal(s)).

which process is in BLOCK state? _____A_____

A none

B P1

C P2

D P3

(3) When designing a disk scheduling policy, which factor has the most significant impact on the disk I/O performance? _____A_____

A. Latency of moving the disk head

B. Time required to read one sector from the disk

C. The average rotational delay

D. The maximal rotational delay

(4) A UNIX file system has 4-KB blocks and 4-byte disk addresses. The inode of file 'X' requires 10 direct entries and one indirect block which is half full. How large is the file in blocks? C

A. 18

B. 266

C. 522

D. 1034

(5) Among following scheduling types, which pair are mainly used to adjust the degree of multiprogramming? A

A. Long-term scheduling, and mid-term scheduling
B. Long-term scheduling, and short-term scheduling
C. Short-term scheduling, and mid-term scheduling
D. Short-term scheduling, and disk scheduling

(6) Which layer of I/O software issues a sequence of commands to the device controller? C

A. User-space I/O software
B. Device Independent I/O software
C. Device driver
D. Interrupt handler

(7) In a pure user-level threads (ULTs) system, there is a process with 3 threads. Which one of the following statements is correct? C

A. In a multiprocessor system, the 3 ULTs may be running at the same time
B. When one ULT in the process is in running state, it is certain that the process is holding a processor
C. When the process is in ready state, one thread may be in running state
D. When one thread issues a blocked system call, another thread in the process may be scheduled to run.

(8) A Linux file whose owner has UID = 12 and GID = 1 has mode rwxr-x---. Tom with UID = 6, GID = 1 tries to do some operations for the file, which one of the following operations is feasible? B

A. Tom can read/write/execute the file.
B. Tom can read the file, but can not write the file.
C. Tom cannot execute any of the read/write/execute operation.
D. Tom can write the file, but not read/execute the file.

(9) A computer has 6 tape drives, with n processes competing for them. Each process may need two drives. For which maximal values of n is the system deadlock free? C

A. 3
B. 4
C. 5
D. 6

(10) Which one of the following actions is effective when thrashing is happening in a computer system? A

A. Destroy some processes
B. Increase the size of disk swapping space
C. Increase the priorities of user processes
D. Upgrade the system with a high-speed CPU

(11) Which one of the following scheduling algorithms is non-preemptive and does NOT suffer from starvation? B

A. Static Priority-based

B. Highest Response Ratio Next

C. Round Robin

D. Shortest Process Next(SJF)

(12) Which one of the following events may cause the running process to be put back into the ready queue? A

A. An interrupt from I/O devices has occurred.

B. The process execute a wait() on a semaphore

C. The process tries to get a message from another process by invoking receive().

D. The process is accessing a page which is not resident in main memory.

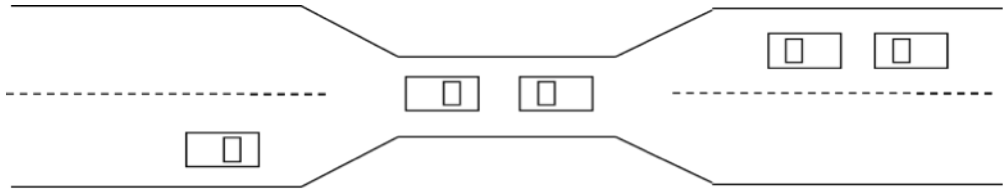
2. Short Answer (25%)

(1) Describe the concept of working set briefly, and explain why it can be used to control the degree of multiprogramming?

Answer:

At any instant of time t, the set of all the pages used by the k most recent memory references is call the working set w(k, t). It works as: the working set of each process is monitored; periodically remove from the resident set of a process those pages that are not in its working set. Because a process may execute only if its working set is in main memory, the control of the degree of multiprogramming is implied by the work set method

(2) A one-way bridge that connect two islands. If two cars from different sides meet on the bridge, none of them is willing to drive the car backward (illustrated as the following diagram).



Please interpret this situation in terms of deadlock, and conclude the necessary and sufficient conditions. What is the typical solution to the problem by a policeman, and what is the risk if the policeman has its own preference.

Ref. Answer:

- Mutual exclusion: one-way bridge
- Hold and wait: two ends as resources
- Non-preemptive: none of the cars is willing to drive the car backward
- Circular waiting: two cars from different sides on the bridge
- The policeman forces one car to drive backward, but if he prefers one direction, starvation risk exists.

(3) Suppose a short-term scheduling algorithm favors those processes that have used little processor time in the recent past. (a) Explain why this algorithm favors I/O-bound processes. (b) Explain

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why this algorithm does not make CPU-bound processes starvation.

Ref. Answer:

- I/O-bound processes use little processor time; thus, the algorithm will favor I/O-bound processes.
- If CPU-bound process is denied access to the processor
  ==> the CPU-bound process won't use the processor in the recent past.
  ==> the CPU-bound process won't be permanently denied access.

(4) Suppose a file contains 10 disk blocks. Now a user process need read the entire file into memory for analysis. Suppose one I/O buffer has the same size as one disk block. It will take OS 100ms to read one disk block into buffer, and 50ms to transfer the data block from buffer to user process. The user process needs 50ms to finish analyzing one data block. What is total time needed for reading and analyzing the entire file if single buffer is used? What's the total time if double buffer is used? Why?

Answer: 1550，1100.

(5) Some file systems allow disk storage to be allocated at different levels of granularity. For instance, a file system could allocate 4 KB of disk space as a single 4-KB block or as eight 512-byte blocks. How could we take advantage of this flexibility to improve performance? What modifications would have to be made to the free-space management scheme in order to support this feature?

Answer: Such a scheme would decrease internal fragmentation. If a file is 5KB, then it could be allocated a 4KB block and two contiguous 512-byte blocks. In addition to maintaining a bitmap of free blocks, one would also have to maintain extra state regarding which of the subblocks are currently being used inside a block. The allocator would then have to examine this extra state to allocate subblocks and coalesce the subblocks to obtain the larger block when all of the subblocks become free.

3. Calculation (22%)

(1) A system has four processes (A, B, C, and D) and five allocatable resources. The current allocation and maximum needs are as follows:

	Allocated	**Maximum**	**Available**
Process A	1 0 2 1 1	1 1 2 1 3	0 0 x 1 2
Process B	2 0 1 1 0	2 2 2 1 0	
Process C	1 1 0 1 0	2 1 3 1 0	
Process D	1 1 1 1 0	1 1 2 2 1	

According to the Banker's algorithm, discuss that

a) What is the smallest value of *x* for which this is a safe state? (4 points)

b) What is the safe sequence? (4 points)

a) x=1

b) D A C B

(2) A computer has a cache, main memory, and a disk used for virtual memory. If a referenced word is in cache, 20ns are required to access it. If it is in main memory but not in the cache, 60ns are needed to load it into the cache, and then the reference is started again. If the word is not in main memory, 12ms are required to fetch the word from disk, followed by 60ns to copy it to the cache, and then the reference is started again. The cache hit ratio is 0.9 and the main memory hit ratio is 0.6. What is the average time, in nanoseconds, required to access a referenced word on this system? (5 points)

Answer: 0.9*20+0.1*(0.6*(60+20)+0.4*(12000000+60+20))=480026 (ns)

(3) In pure on-demand paging, a page replacement policy is used to manage system resources. Suppose that a newly-created process has 3 page frames allocated to it, and then generates the following string of page references :

A B C B A D A B C D A B A C B D

(i) How many page faults would occur with FIFO page replacement?_____

A B C B A D A B C D A B A C B D

(ii) How many page faults would occur with LRU page replacement?_____

A B C B A D A B C D A B A C B D

(iii) How many page faults would occur with OPTIMAL page replacement?_____

A B C B A D A B C D A B A C B D

NOTE: Please indicate which page references have led to those page faults by drawing a circle surround them. (9 points)

Answer:

(i) How many page faults would occur with **FIFO** page replacement? 12

A B C B A **D A B C D A B** A **C** B **D**

The **bold font** indicates the references that cause a page fault.

(ii) How many page faults would occur with **LRU** page replacement? 10

A B C B A **D** A B **C D A B** A **C** B **D**

The **bold font** indicates the references that cause a page fault.

(iii) How many page faults would occur with **OPT** page replacement? 7

A B C B A **D** A B **C** D A **B** A C B **D**

The **bold font** indicates the references that cause a page fault.

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4. Analysis (17%)

(1) How would use of a RAM disk affect your selection of a disk-scheduling algorithm? What factors would you need to consider? Do the same considerations apply to hard-disk scheduling, given that the file system stores recently used blocks in a buffer cache in main memory? (5%)

Answer: Disk scheduling attempts to reduce the overhead time of disk head positioning. Since a RAM disk has uniform access times, scheduling is largely unnecessary. The comparison between RAM disk and the main memory disk-cache has no implications for hard-disk scheduling because we schedule only the buffer cache misses, not the requests that find their data in main memory.

(2) The students like accessing campus BBS. Two engineers maintain the BBS, one for software maintenance and the other for hardware repairing. The following policies are enforced:

- Students can access the BBS when neither of the engineers are maintaining the system;
- An engineer can not start maintenance when some students are using BBS;
- Hardware and software maintenance cannot be in action at the same time.
- Students are not allowed to access BBS during maintenance.

Try to simulate the scenario using pseudo-code in terms of semaphore. (12 points)

Ref. Answer:

This scenario can be modeled as the reader/writer problem that the readers have priority.

Pseudo-code is:

```
semaphore rSem = 1;      /* 1 point */
semaphore wSem = 1;     /* 1 point */
int readCount = 0;

void students()
{
while(TRUE)    /* 0.5 point */
{
      wait(rSem);     /* Mutex, 2 points */
            readCount++;
            if (readCount == 1)
                  wait(wSem);    /* 1 point */
      signal(rSem);
      Surf();
      wait(rSem);     /* Mutex, 2 points */
            readCount--;
            if (readCount == 0)
                  signal(wSem);    /* 1 point */
      signal(rSem);
}
}
```

```
void engineers()
{
while(TRUE) /* 0.5point */
{
      wait(wSem);      /* Mutex, 2 points */
            Maintain();
      signal(wSem);
}
}

void main()
{
parbegin(students, engineers);/* 1 point*/
}
```

（装订线内不要答题）